Derivation appendix: 09 — the residual $?_2$ and the seesaw neutrino sector

the residual Z_2 selection rule, and the seesaw neutrino sector

1. The residual $\mathbb{Z}_2$ of the aligned vacuum is the S_3 reflection ab about the vacuum direction

derived here — §1

Independent checks

- found by testing all six S_3 elements on the aligned vacuum: exactly one non-identity element fixes it
- the potential is invariant under it (direct substitution)
- each soft pair splits into one $\mathbb{Z}_2$-even and one $\mathbb{Z}_2$-odd combination

Would be falsified by: a $\mathbb{Z}_2$-odd soft combination passing the invariance check (falsify run)

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2. Parities of the physical states: e, h_0, one A and one $H^\pm$ are odd; μ, τ and the other four scalars even

derived here — §2

Independent checks

- rotation columns tested as G-eigenvectors: (vacuum, in-plane, out-of-plane) = (+, ?, +)
- $G^\mathsf{T} M_\ell G = M_\ell$ symbolically; the odd eigenvalue equals m_e at a fitted point to 10^{-9}

Would be falsified by: a mass eigenstate that is not a G-eigenvector (z2.parity raises)

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3. Selection rule: an amplitude whose external states multiply to -1 vanishes to all orders

derived here — §3–4

Claim

$$\prod_i P_i = -1 \;\Rightarrow\; \mathcal{A} = 0: \quad \mu \to e\gamma,\ \tau \to e\gamma,\ \mu \to 3e,\ \mu N \to eN,\ h_{\text{even}} \to e\ell,\ h_{\text{odd}} \to \tau\mu \text{ forbidden}$$

Independent checks

- every odd-parity scalar-lepton coupling $<$ 1e-14 over 594 exact (point, mu3) evaluations, parities read per eigenvector
- allowed off-diagonal couplings all non-zero (the test is not vacuous)
- notebook 08's B(tau -> e gamma) is round-off at all 118 380 exact evaluations

Would be falsified by: a non-zero coupling with odd total parity at any exact point

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4. Exact S_3 forces a single non-zero entry in the PMNS electron row: no solar or reactor mixing

derived here — §5–6

Claim

$$G^{\mathsf{T}} m_D G = m_D,\ G^{\mathsf{T}} M_R G = M_R \ \Rightarrow\ [G, m_\nu] = 0 \ \Rightarrow\ |U_{ei}| = \delta_{i,\mathrm{odd}}$$

Independent checks

G, m_nu = 0 symbolically, for all five Yukawas and M, M_S

- electron row single-entry to 1e-12 for 200 random S3-invariant sectors
- the lepton analogue of V_us = 0 at exact S3 (notebook 02, finding 9)

Would be falsified by: one Z2-odd Majorana entry gives two non-zero electron-row entries (falsify run)

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5. NuFIT 6.0 / PDG oscillation inputs (normal ordering)

imported — §7

Independent checks

- θ_{12}, $\sin^2\theta_{13}$, θ_{23} and Δm^2_{21} checked against web-search excerpts of [NuFIT6] and [PDG]
- $\Delta m^2_{31} = 2.5 \times 10^{-3}$ eV^2 is an order-of-magnitude input, not a quoted best fit

Would be falsified by: the published Table 1 differing from these inputs beyond their uncertainties

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6. A $\mathbb{Z}_2$-odd soft breaking of M_R alone reproduces the oscillation data

derived here — §7–8

Independent checks

- exact fit of 5 oscillation observables with m1 = 1 meV, from 80 random starts at M = 1e14 GeV
- a penalized search keeps the fit exact down to |eps|/M = 5e-4
- only the two Z2-odd Majorana directions are switched on; Yukawas, potential and vacuum untouched
- with them off the electron row is single-entry (section 6), so they are necessary

Would be falsified by: no exact fit with only the Z2-odd Majorana directions

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7. Charged-scalar dipole function for a neutral fermion of any mass

derived here — §9

Claim

$$f(t) = \int_0^1 \frac{x^2(1-x)}{x + (1-x)t}\,dx = \frac{1 - 6t + 3t^2 + 2t^3 - 6t^2 \ln t}{6(1-t)^4}$$

Independent checks

- closed form = Feynman-parameter integral to 1e-8
- f(0) reproduces radiative.one_loop_charged exactly
- f -> 1/(3t): a heavy neutrino decouples

Would be falsified by: a massless limit differing from notebook 07's validated charged loop

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8. The M_R breaking leaks into $\mu \to e\gamma$ only through m_ν/M_R-suppressed neutrino loops

derived here — §9

Independent checks

- H+-N loops computed on the fit benchmark (point 0 of notebook 01, mu3 = 1 GeV) for M = 1e3 ... 1e14 GeV
- W-N loops bounded with an O(1) loop function
- generic fit: both below 1e-30; the smallest-breaking benchmark (mixing enhanced ~1e3 by a cancellation in m_nu): below 1e-19
- MEG II limit 1.5e-13

Would be falsified by: a seesaw scale at which either piece reaches the MEG II sensitivity

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